

# Audit Management System – Microservices Architecture Document

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## 1. Overview

This document defines the microservices architecture for the Audit Management System based on the BRD. It includes:

- Service boundaries
  - Model allocation
  - Improved data models
  - Field-level descriptions
  - Design improvements for scalability
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## 2. Microservices Overview

The system is divided into 8 microservices: 1. User Service 2. Audit Service 3. Observation Service 4. Action Service 5. Approval Service 6. File Service 7. Notification Service 8. Reporting Service

Each service:

- Owns its database
  - Has independent deployment
  - Communicates via APIs/events
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## 3. USER SERVICE

### Responsibility

- Authentication & Authorization
- Role management
- Organizational structure

## Models

### User

| Field        | Type      | Description                      |
|--------------|-----------|----------------------------------|
| UserId       | int       | Unique identifier                |
| Name         | string    | Full name                        |
| Email        | string    | Unique email                     |
| PasswordHash | string    | Hashed password (NOT plain text) |
| RoleId       | int       | Role reference                   |
| DepartmentId | int?      | Department reference             |
| Expertise    | string?   | Skill/area                       |
| CreatedAt    | datetime  | Creation timestamp               |
| UpdatedAt    | datetime? | Last update                      |
| IsDeleted    | bool      | Soft delete flag                 |

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### Role

| Field    | Type   | Description       |
|----------|--------|-------------------|
| RoleId   | int    | Unique identifier |
| RoleName | string | Admin/Auditor/etc |

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### Department

| Field          | Type     | Description       |
|----------------|----------|-------------------|
| DepartmentId   | int      | Unique identifier |
| DepartmentName | string   | Name              |
| CreatedAt      | datetime | Created timestamp |

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## 4. AUDIT SERVICE

### Responsibility

- Audit lifecycle
- Scheduling
- Auditor assignment

### Models

#### Audit

| Field           | Type        | Description                  |
|-----------------|-------------|------------------------------|
| AuditId         | int         | Unique ID                    |
| AuditCode       | string      | Unique code                  |
| AuditName       | string      | Name                         |
| DepartmentId    | int         | Reference to department      |
| AuditType       | enum        | Internal/External/Compliance |
| CreatedByUserId | int         | Creator reference            |
| StartDate       | date        | Planned start                |
| EndDate         | date        | Planned end                  |
| Status          | AuditStatus | Enum status                  |
| CreatedAt       | datetime    | Created time                 |
| UpdatedAt       | datetime?   | Updated time                 |
| IsDeleted       | bool        | Soft delete                  |

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#### AuditAuditor

| Field   | Type | Description       |
|---------|------|-------------------|
| Id      | int  | Unique ID         |
| AuditId | int  | Audit reference   |
| UserId  | int  | Auditor reference |

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## AuditHistory

| Field           | Type     | Description     |
|-----------------|----------|-----------------|
| Id              | int      | Unique ID       |
| AuditId         | int      | Audit reference |
| FieldChanged    | string   | Changed field   |
| OldValue        | string   | Previous value  |
| NewValue        | string   | Updated value   |
| ChangedByUserId | int      | Who changed     |
| ChangedAt       | datetime | Timestamp       |

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## 5. OBSERVATION SERVICE

### Responsibility

- Capture audit findings
- Manage severity and submission

### Models

#### Observation

| Field           | Type     | Description              |
|-----------------|----------|--------------------------|
| ObservationId   | int      | Unique ID                |
| AuditId         | int      | Related audit            |
| Title           | string   | Observation title        |
| Description     | string   | Details                  |
| Severity        | enum     | Low/Medium/High/Critical |
| PolicyReference | string   | Optional reference       |
| Status          | enum     | Draft/Submitted/Approved |
| DueDate         | date     | Due date                 |
| CreatedAt       | datetime | Created                  |
| UpdatedAt       | datetime | Updated                  |

| Field     | Type | Description |
|-----------|------|-------------|
| IsDeleted | bool | Soft delete |

## ObservationHistory

| Field         | Type     | Description |
|---------------|----------|-------------|
| Id            | int      | Unique ID   |
| ObservationId | int      | Reference   |
| FieldChanged  | string   | Field name  |
| OldValue      | string   | Old value   |
| NewValue      | string   | New value   |
| ChangedAt     | datetime | Timestamp   |

# 6. ACTION SERVICE

## Responsibility

- Manage corrective actions
- Track progress and closure

## Models

### CorrectiveAction

| Field             | Type         | Description        |
|-------------------|--------------|--------------------|
| ActionId          | int          | Unique ID          |
| ObservationId     | int          | Linked observation |
| AssignedToUserId  | int          | Responsible user   |
| ActionDescription | string       | Description        |
| RootCause         | string       | Cause              |
| ExpectedOutcome   | string       | Outcome            |
| DueDate           | date         | Deadline           |
| Status            | ActionStatus | Enum               |

| Field     | Type     | Description |
|-----------|----------|-------------|
| CreatedAt | datetime | Created     |
| UpdatedAt | datetime | Updated     |
| IsDeleted | bool     | Soft delete |

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## ActionHistory

| Field        | Type     | Description |
|--------------|----------|-------------|
| Id           | int      | Unique ID   |
| ActionId     | int      | Reference   |
| FieldChanged | string   | Field       |
| OldValue     | string   | Old         |
| NewValue     | string   | New         |
| ChangedAt    | datetime | Timestamp   |

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# 7. APPROVAL SERVICE

## Responsibility

- Central approval engine
- Handles all approval workflows

## Models

### ApprovalRequest

| Field             | Type | Description               |
|-------------------|------|---------------------------|
| ApprovalId        | int  | Unique ID                 |
| EntityType        | enum | Audit/Observation/Action  |
| EntityId          | int  | Entity reference          |
| ApprovalType      | enum | Type of approval          |
| RequestedByUserId | int  | Requester                 |
| Status            | enum | Pending/Approved/Rejected |

| Field     | Type     | Description |
|-----------|----------|-------------|
| CreatedAt | datetime | Created     |
| UpdatedAt | datetime | Updated     |

## ApprovalHistory

| Field          | Type     | Description       |
|----------------|----------|-------------------|
| HistoryId      | int      | Unique ID         |
| ApprovalId     | int      | Reference         |
| ActionByUserId | int      | Actor             |
| Action         | enum     | Approved/Rejected |
| Comments       | string   | Optional comments |
| ActionDate     | datetime | Timestamp         |

# 8. FILE SERVICE

## Responsibility

- Manage evidence and attachments

## Models

### FileAttachment

| Field            | Type     | Description              |
|------------------|----------|--------------------------|
| FileId           | int      | Unique ID                |
| EntityType       | enum     | Audit/Observation/Action |
| EntityId         | int      | Linked entity            |
| FilePath         | string   | Storage path             |
| UploadedByUserId | int      | Uploader                 |
| CreatedAt        | datetime | Timestamp                |

## 9. NOTIFICATION SERVICE

### Responsibility

- Send alerts and system notifications

### Models

#### Notification

| Field          | Type     | Description |
|----------------|----------|-------------|
| NotificationId | int      | Unique ID   |
| UserId         | int      | Target user |
| Message        | string   | Content     |
| IsRead         | bool     | Read status |
| CreatedAt      | datetime | Timestamp   |

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#### NotificationTemplate

| Field      | Type   | Description   |
|------------|--------|---------------|
| TemplateId | int    | Unique ID     |
| Name       | string | Template name |
| Content    | string | Template body |

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## 10. REPORTING SERVICE

### Responsibility

- Read-only analytics
- Dashboard data aggregation

### Models

- No primary database models
- Uses read replicas / API aggregation



## 11. KEY DESIGN IMPROVEMENTS

- Use ENUMS instead of strings
  - No cross-service foreign keys
  - Add history tables for audit trail
  - Store only reference IDs between services
  - Use event-driven communication
  - Secure passwords with hashing
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## 12. EVENT FLOW (RECOMMENDED)

- AuditCreated
  - AuditApproved
  - ObservationSubmitted
  - ActionAssigned
  - ActionClosed
  - ApprovalCompleted
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## 13. CONCLUSION

This architecture ensures:

- Scalability
  - Maintainability
  - Clear service ownership
  - Alignment with BRD requirements
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